Exercise Sheet 2: Operations with Matrices

October 2024

1 Matrices and Gaussian Elimination

Operations on matrices

Recall:

1) Some special matrices: Identity matrices, zero matrices, symmetric matrices, ...

2) Operations: Addition, multiplication with a scalar, transpose of a matrix, multiplication.

3) General properties: Associativity, commutativity, ...

Exercise 1. Suppose that A, B, C, D, E are matrices with the following sizes: $A : 4 \times 5$; $B : 4 \times 5$; $C : 5 \times 2$; $D : 4 \times 2$; $E : 5 \times 5$. Determine (if it exists) the size of each of the following matrices:

1. BA;
2. $AC + D$;
3. $AE + B$;
4. $AB + B$;
5. $2E(A + B)$;
6. $E(AC)$;
7. $E^T A$;
8. $(A^T + E)D$.

Exercise 2. Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}, C = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}, D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}, E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}.$$

Compute the following (where possible):

1. $2B - C$;
2. $3D - 2E^T$;
3. $3D^T - 2E$ and $Tr(3D - 2E^T)$;
4. AB and BA;
5. AC and CA;
6. $(C^T B)A^T$ and $Tr((C^T B)A^T)$;
7. $Tr(DD^T)$;
8. $D^T E^T - (ED)^T$.

Exercise 3. Write down the 2 by 2 matrices A and B that have entries $a_{ij} = i + j$ and $b_{ij} = (-1)^{i+j}$. Multiply them to find AB and BA. Is the product of A and B commutative?

Exercise 4. Which of the following matrices are guaranteed to equal $(A + B)^2$?

1. $A^2 + 2AB + B^2$
2. $A(A + B) + B(A + B)$
3. $(A + B)(B + A)$
4. $A^2 + AB + BA + B^2$

Exercise 5. Suppose A commutes with every 2 by 2 matrix, and in particular

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ commutes with } B_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Show that $a = d$ and $b = c = 0$. Consequently, prove that if $AB = BA$ for all 2×2-matrices B, then A is a multiple of the identity.

Exercise 6. In each part, find matrices A, X, B which express the given system of linear equations as a single matrix equation $AX = B$. Solve these equations.

1.

$$\begin{aligned} x_1 - 3x_2 + 5x_3 &= 7 \\ 9x_1 - x_2 \quad + x_3 &= -1 \\ x_1 + 5x_2 + 4x_3 &= 0 \end{aligned}$$

2.

$$\begin{aligned} x_1 \qquad - 3x_3 + x_4 \qquad &= 7 \\ 5x_1 + x_2 \qquad - 8x_4 \qquad &= 3 \\ 2x_1 - 5x_2 + 9x_3 - x_4 \qquad &= 0 \\ 3x_2 \quad - x_3 + 7x_4 &= 2 \end{aligned}$$

Exercise 7. Find the powers A^2, A^3, B^2, B^3, C^2, C^3. What are A^k, B^k and C^k for a given k?

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \text{ and } C = AB = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

Inverses of matrices

Recall:

1) Equivalent definitions: $A \in M_{n\times n}$ is invertible if one the following equivalent conditions holds.
   - i) There exists $B \in M_{n\times n}$ such that $AB = BA = I$;
   - ii) The equation $Ax = b$ has a unique solution for every $b \in M_{n\times 1}$;
   - ii) The equation $Ax = 0$ has only the trivial solution, i.e., $x = 0$.
2) Find the inverse of a matrix.
   - Using elementary row transformations to bring $[A|I]$ into $[I|A^{-1}]$. (Gauss–Jordan)

Exercise 8. Use the Gauss–Jordan method to invert the following matrices then solve the equations $Ax = b$ for $b = \begin{bmatrix} -1 \\ 2 \\ 7 \end{bmatrix}$.

1.

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

2.

$$A_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix},$$

3.

$$A_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix},$$

4.

$$A_4 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix},$$

5.

$$A_5 = \begin{bmatrix} \sqrt{2} & 3\sqrt{2} & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

6.

$$A_6 = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & \frac{-2}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{10} \\ \frac{1}{5} & \frac{-4}{5} & \frac{1}{10} \end{bmatrix}.$$

Exercise 9. Find the inverse (in any legal way) of

1.

$$A_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix},$$

2.

$$A_2 = \begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix},$$

3.

$$A_3 = \begin{bmatrix} k & 0 & 0 & 0 \\ 1 & k & 0 & 0 \\ 0 & 1 & k & 0 \\ 0 & 0 & 1 & k \end{bmatrix}.$$

Exercise 10. For which numbers c is this matrix not invertible, and why not?

$$A = \begin{bmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{bmatrix}$$

Exercise 11. Give examples of matrices A and B such that

(a) $A + B$ is not invertible although A and B are invertible.

(b) $A + B$ is invertible although A and B are not invertible.

(c) All of A, B, and $A + B$ are invertible.

(d) In the last case, use $A^{-1}(A+B)B^{-1} = B^{-1} + A^{-1}$ to show that $C = B^{-1} + A^{-1}$ is also invertible and find a formula for C.

Exercise 12. Show that $A^2 = 0$ is possible but $A^T A = 0$ is not possible (unless A is the zero matrix).

Exercise 13. If the inverse of A^2 is B, show that the inverse of A is AB. Thus, A is invertible whenever A^2 is invertible.

Exercise 14. If $A = A^T$ and $B = B^T$, which of the following matrices are certainly symmetric?

1. $A^2 - B^2$;
2. $(A + B)(A - B)$;
3. ABA;
4. $ABAB$.